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WAN Operations

7.2.1 WAN Standards

Note that you understand how critical WANs are to large networks, this topic discusses how they work. The concept of a WAN is defined by the OSI model and consists of many different standards and protocols. Cisco routers, switches, firewalls, servers, systems, terminals, and now data networks. Many of the technologies and standards developed for these WANs were used as the basis for network WANs.

Modern WAN standards are defined and managed by a number of recognized authorities including the following:

- **TIA/EIA** – Telecommunications Industry Association and Electronic Industries Alliance
- **ISO** – International Organization for Standardization
- **IEEE** – Institute of Electrical and Electronics Engineers

Both standards are essentially the same standards and their transmission capacity can be increased by using DWDM technology.

Layer 1 Protocols

Layer 1 protocols describe the electrical, mechanical, and operational components needed to transmit bits over a WAN. For example, service providers commonly use high-bandwidth optical fiber media to carry long-distance (i.e., long haul) using the following Layer 1 optical fiber protocol standards:

- Synchronous Digital Hierarchy (SDH)
- Synchronous Optical Networking (SONET)
- Dense Wavelength Division Multiplexing (DWDM)

SDH and SONET essentially provide the same services and their transmission capacity can be increased by using DWDM technology.

Layer 2 Protocols

Layer 2 protocols define how data will be encapsulated into a frame.

Several Layer 2 protocols have evolved over the years including the following:

- Broadband (i.e., DSL and Cable)
- Wireless
- Ethernet WAN (Metro Ethernet)
- Multiprotocol Label Switching (MPLS)
- Point-to-Point Protocol (PPP) (less used)
- High-Level Data Link Control (HDLC) (less used)
- Frame Relay (legacy)
- Asynchronous Transfer Mode (ATM) (legacy)

7.2.2 WANs in the OSI Model

Most WAN standards focus on the physical layer (OSI Layer 1) and the data link layer (OSI Layer 2), as shown in the figure.

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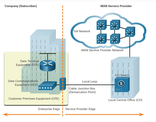
7.2.3 Common WAN Terminology

The WAN physical layer describes the physical connections between the company network and the service provider network.

There are specific terms used to describe WAN connections between the subscriber (i.e., the company / client) and the WAN service provider, as shown in the figure.

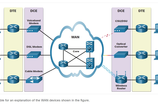


Refer to the table for an explanation of the term shown in the figure, as well as some additional WAN related terms.

WAN Term	Description
Data Terminal Equipment (DTE)	• This is the device that connects the subscriber LAN to the WAN communication device (i.e., DCE). • Most routers are DTE devices by default. • The DTE connects to the local loop through the DCE. • The DTE device is usually a router but could be a host or server.
Data Communications Equipment (DCE)	• Also called data circuit-terminating equipment, this is the device used to communicate with the provider. • The DCE primarily provides an interface to connect subscribers to a communication link on the WAN cloud.
Customer Premises Equipment (CPE)	• This is the DTE and DCE devices (i.e., router, modem, optical converter) located at the subscriber's premises. • The subscriber either owns the CPE or leases the CPE from the service provider.
Point-of-Presence (POP)	• This is the point where the subscriber connects to the service provider network.
Demarcation Point	• This is a physical location in a building or complex that officially separates the customer premises from the provider network. • The demarcation point is typically a cabling junction box, located on the customer premises, that connects the CPE wiring to the local loop. • It is important because it is the point where responsibility changes from the subscriber to the service provider. • When problems arise, it is necessary to determine whether the user or the service provider is responsible for troubleshooting or repair.
Local Loop (or last mile)	• This is the actual copper or fiber cable that connects the CPE to the CO of the service provider.
Central Office (CO)	• This is the local service provider facility or building that connects the CPE to the provider network.
Toll network	• This includes backhaul, long haul, all-digital, fiber-optic communications lines, switches, routers, and other equipment inside the WAN provider network.
Backbone network	• Well-ordered, fiber/wireless network connect multiple access nodes of the service provider network. • Backbone networks can span over municipalities, countries and regions. • Backbone networks are also connected to Internet service providers and to the backbone network.
Backbone network	• Well-ordered These are large, high-capacity networks used to interconnect service provider networks and to create a redundant network. • Backbone network can be owned by a single service provider, through another service provider. • Backbone network service providers are also called Tier-1 providers.

7.2.4 WAN Devices

There are many types of devices that are specific to WAN environments. However, the end to end data path over a WAN is mostly from source DTE to the DCE, then to the WAN cloud, then to the DCE to, and finally to the destination DTE, as shown in the figure.



Refer to table for an explanation of the WAN devices shown in the figure.

WAN Device	Description
Voiceband Modem	• Converts analog or dial up modem. • Legacy device that converted (i.e., modulated) the digital signals produced by a computer into analog voice frequencies. • Uses telephone lines to transmit data.
DSL Modem and **Cable Modem**	• Collectively known as broadband modems, these high-speed digital modems connect to the DTE router using Ethernet. • DSL modems connect to the WAN using telephone lines. • Cable modems connect to the WAN using cable lines. • Both operate in a similar manner to the aforementioned modem but use higher broadband frequencies and transmission options.
CSU/DSU	• Digital leased lines require a CSU and a DSU. • It connects a digital device to a digital line. • This device can be a separate device like a modem or it can be an interface on a router. • The CSU provides termination for the digital signal and ensures connection integrity through error correction and line monitoring. • The DSU converts the line frames into frames that the LAN can interpret and vice versa.
Optical Converter	• Also known as an optical fiber converter. • These devices connect fiber optic media to copper media and convert optical signals to electronic pulses.
Wireless Router or **Access Point**	• Devices are used to wirelessly connect to a WAN provider. • Routers could also use cellular wireless connectivity.
WAN Core devices	• The WAN backbone consists of multiple high-speed routers and Layer 3 switches. • A router or multilayer switch must be able to support multiple telecommunications interfaces of speeds and a large amount of RAM to hold routing tables. • A host also be linked IP packets at full speed on all these interfaces. • The router or multilayer switch must also support the routing protocols being used in the core.

Note: The preceding list is not exhaustive and other devices may be required, depending on the WAN access technology chosen.

7.2.5 Serial Communication

Almost all network communications occur using a serial communication delivery. Serial communication transmits bits sequentially over a single channel. In contrast, parallel communication simultaneously transmit several bits using multiple wires.

Click Play to see an illustration of the difference between serial and parallel connections.

Although a parallel connection theoretically transfers data eight times faster than a serial connection, it is prone to synchronization problems. As the cable length increases, the synchronization between multiple channels becomes more sensitive to distance. For this reason, parallel communication is limited to very short distances only (e.g., copper media is limited to less than 5 meters (i.e., 16 feet).

Therefore, parallel communication is not a viable WAN communication method because of its length restriction. It is however a viable solution in data centers where distances between servers and switches are relatively short.

For instance, the Cisco Nexus switches in data centers support parallel optics solutions to transfer more data signals and achieve higher speeds (i.e., 40 Gbps and 100 Gbps).

7.2.6 Circuit-Switched Communication

Network communication can be implemented using circuit-switched communication. A circuit-switched network establishes a dedicated circuit (or channel) between endpoints before the users can communicate.

Specifically, circuit switching dynamically establishes a dedicated virtual connection through the service provider network before voice or data communication can start.

For example, when a user makes a telephone call using a landline, the number called is used by the provider equipment to create a dedicated circuit from the caller to the called party.

Note: A landline describes a telephone situated in a specific location that is connected to the provider using copper or fiber optic media.

During transmission over a circuit-switched network, all communication uses the same path. The entire fixed capacity allocated to the circuit is available for the duration of the connection, regardless of whether there is information to transmit or not. This can be inefficient because other users cannot share the circuit, even if something is generally not wanted to be sent/communicated.

The two most common types of circuit-switched WAN technologies are the public switched telephone network (PSTN) and the legacy Integrated Services Digital Network (ISDN).

Click Play in the figure to see how circuit switching works.

7.2.7 Packet-Switched Communications

Network communication is most commonly implemented using packet-switched communication. In contrast to circuit-switching, packet switching does not require a dedicated circuit to be established. Instead, packet switching splits traffic into packets that require a circuit to be established, and they share many pairs of nodes to communicate over the same channel.

Packet switching is much less expensive and more flexible than circuit switching. Although susceptible to delay (latency) and variability of delay (jitter), packet switching delivers satisfactory transport of voice and video communications or data traffic.

Common types of packet-switched WAN technologies are Ethernet WAN (Metro Ethernet), Multiprotocol Label Switching (MPLS), as well as legacy Frame Relay and legacy Asynchronous Transfer Mode (ATM).

Click Play in the figure to see a packet switching example.

7.2.8 SDH, SONET, and DWDM

Service provider networks use fiber-optic infrastructures to transport user data between destinations. Fiber optic cable is far superior to copper cable for long distance transmission due to its much lower attenuation and interference.

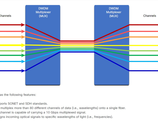
There are two optical fiber OSI layer 1 standards available to service providers:

- **SDH** – Synchronous Digital Hierarchy (SDH) is a global standard for transporting data over fiber optic cable.
- **SONET** – Synchronous Optical Networking (SONET) is the North American version of SDH and provides the same services as SDH.

Both standards are essentially the same and therefore, they are often listed as SONET/SDH.

SDH/SONET define how to transfer multiple data, voice, and video communications over optical fiber using lasers or light-emitting diodes (LEDs) over long distances. Small variations are used in the ring network topology that contains the redundant fiber paths that allow traffic to flow in both directions.

Dense Wavelength Division Multiplexing (DWDM) is a newer technology that increases the data carrying capacity of SDH and SONET by simultaneously sending multiple streams of data (multiplexing) using different wavelengths of light, as shown in the figure.



DWDM has the following features:

- It supports SONET and SDH standards.
- It multiplexes up to 32 SONET/SDH data streams of data (i.e., wavelengths) onto a single fiber.
- Each channel is capable of carrying a 10 Gbps multiplexed signal.
- It assigns incoming optical signals to specific wavelengths of light (i.e., frequencies).

Note: DWDM circuits are used in long haul systems and modern submarine communications cable systems.

7.2.9 Check Your Understanding - WAN Operations

Check your understanding of WAN operations by choosing the BEST answer to the following questions.

1. Which two statements about the WAN OSI Layer 1 are true? (Choose two.)
   - [ ] describes how data will be encapsulated into a frame.
   - [x] describes the electrical, mechanical, and operational components needed to transmit bits.
   - [ ] includes protocols such as PPP, HDLC, and Ethernet.
   - [x] includes protocols such as SDH, SONET, and DWDM.
2. Which WAN term defines the point where the subscriber connects to the service provider network.
   - [ ] Customer Premises Equipment (CPE)
   - [ ] Data Communications Equipment (DCE)
   - [ ] Demarcation point
   - [ ] Local Loop
   - [x] Point-of-Presence (POP)
3. Which two devices operate in a similar manner to the aforementioned modem but use higher broadband frequencies and transmission speeds. (Choose two.)
   - [x] Cable Modem
   - [ ] CSU/DSU
   - [x] DSL Modem
   - [ ] Optical Converter
   - [ ] Voiceband Modem
4. Which communication method is used in all WAN connections?
   - [ ] Circuit-Switched
   - [ ] Packet-Switched
   - [ ] Parallel
   - [x] Serial
5. Which two WAN connectivity options are circuit-switched technologies? (Choose two.)
   - [ ] ATM
   - [ ] Ethernet WAN
   - [ ] Frame Relay
   - [x] ISDN
   - [x] PSTN
6. Which two WAN connectivity options are packet-switched technologies? (Choose two.)
   - [x] Ethernet WAN
   - [x] Frame Relay
   - [ ] ISDN
   - [ ] PSTN
7. Which service provider fiber optic technology increases the data carrying capacity using different wavelengths?
   - [x] DWDM
   - [ ] SDH
   - [ ] SONET

Check | Show Me | Reset